



FEEDBACK

Mini Offline UPS

In 'Mini Offline UPS' DIY article published in March issue, what changes are required in the circuit to use it for 1kVA load? Also, how can the backup time be increased to, say, one hour?

Anirvan K.
Kolkata

The author Zameeruddin Syed replies:

First, for the UPS to support 1kVA load, change the transformer to 1.5kVA rating because transformer efficiency will not be 100 per cent. Then, add more 2N3055 transistors. We have used eight (four + four) transistors for 350VA UPS design. For 1kVA UPS, we have to use 14 (seven + seven) transistors with the remaining circuit as it is.

Battery backup depends on the current consumption of the load and the ampere-hour capacity of the battery. For example, if you are using a 12V, 100Ah battery for the UPS/inverter, and the load you have connected is consuming 10A, then back up will be 100Ah/10A for ten hours. But, this value depends on the condition of the battery, constant current consumption of the load and losses, if any.

Thermoelectric Fridge

I have doubt regarding 'Thermoelectric Refrigerator' DIY project published in March 2014 issue. I have made the circuit as mentioned in the magazine. I connected two modules; on both sides I used water for cooling and also as coolant. The problem is that I am not getting low temperature on the colder side. I have done experiments and the minimum temperature we got was 23°C only. How can I increase efficiency to get a lower temperature?

Jitendra Mishra
Through email

From Facebook: EFY's Electronics Design Community

I need a project that can control home appliances using an Android phone. Please help.

Shahzad Ashraf

EFY: You may refer to 'Home Automation Through Bluetooth' DIY project published in November 2013 issue.

□ I want to make a simple project using ICs. Please help.

Pratheesh Rko

EFY: Pick up any EFY magazine, you will find many simple projects using ICs. You can also refer to *Simple Projects You Can Make At Home* book published by EFY. If interested, you may contact our associates, Kits'n'Spares (info@kitsnspares.com), to get the book or older issues of EFY magazine.

'Spot An Error' Award Winners

In First Look section in April issue, the price of Moto E 2nd Gen smartphone from Motorola is incorrectly mentioned as ₹ 5999; instead it should be ₹ 6999.

In 'Air Motion Detector' DIY in April issue, resistor R6 (22-kilo-ohm) is used in the circuit but is missing in Parts List.

Siddharth Kaul

□ In 'Wireless Hotel Ordering System' DIY in April issue, in PCB layout of the transmitter section, one pin of capacitor C1 (0.1μF) is connected to ground but the other pin is not connected in the PCB.

Rakesh Thakur

□ In 'Wireless Hotel Ordering System' article, in PCB layout, the ATmega8 microcontroller is shown as DIP28, 15.24mm (or 0.6-inch) package. But the ATmega8 microcontroller comes in 32A or 28P3 package.

Janak Nepal

The author Sameer Ghewari replies:
Please mention more details on the power supply you are using to power the modules.

Each module consumes up to 60W (12V, 5A), if kept continuously on. It is possible that the supply is not providing enough current. A quick test is to check if the cold side of the thermocouple becomes cold on touch.

Lack of a heat-sink on the hot side can also be an issue. You need good amount of thermal heat sinking, otherwise it will not cool well.

Power Amplifier

This is regarding 'Power Amplifier for FM Transmitter' DIY project published in November 2014 issue. Please tell me whether this circuit can amplify the available mobile

Errata

In 'Buyers' Guide' on LED bulbs (May issue, pages 108-109), EFY efficiency factor for Bajaj LED bulb should be 85.71 instead of 67.14.

Luminance of Syska LED bulb is 730lm instead of 600lm, therefore EFY lumens factor would be 1.22 and EFY efficiency factor would be 104.29.

MRP of Surya LED bulb is ₹ 400, therefore EFY lumens factor would be 1.40.

network signal. If not, suggest other solutions.

Praveen Reddy Gorla
Kukatpally, Hyderabad

The author Joy Mukherji replies:

The existing mobile signal can be amplified with the help of a GSM 900MHz repeater/booster along with a Yagi antenna. These are widely available on eBay from China.